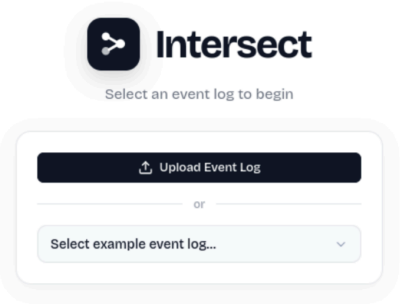
Henri Julius

Intersect. (M1 Prototype)

Finding and analyzing inter-object dependencies in OCELs

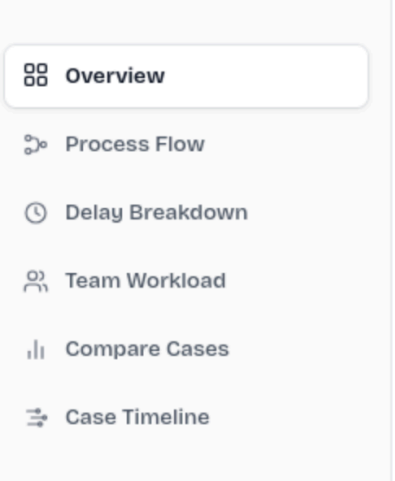
Intersect is a web-based application mainly aimed at non-technical users such as project managers and team leaders. Analysts can use the tool to prepare process-specific textual explanations for delays such that non-technical users can quickly connect a specific delay to the real-world process. The source code for the project can be accessed under *github.com/HenriJul/intersect* and a live version of the prototype can be found at vercel.app.

After opening the website, an event log has to be selected. In this prototype file selection is non-functional and instead one of two example logs can be selected.

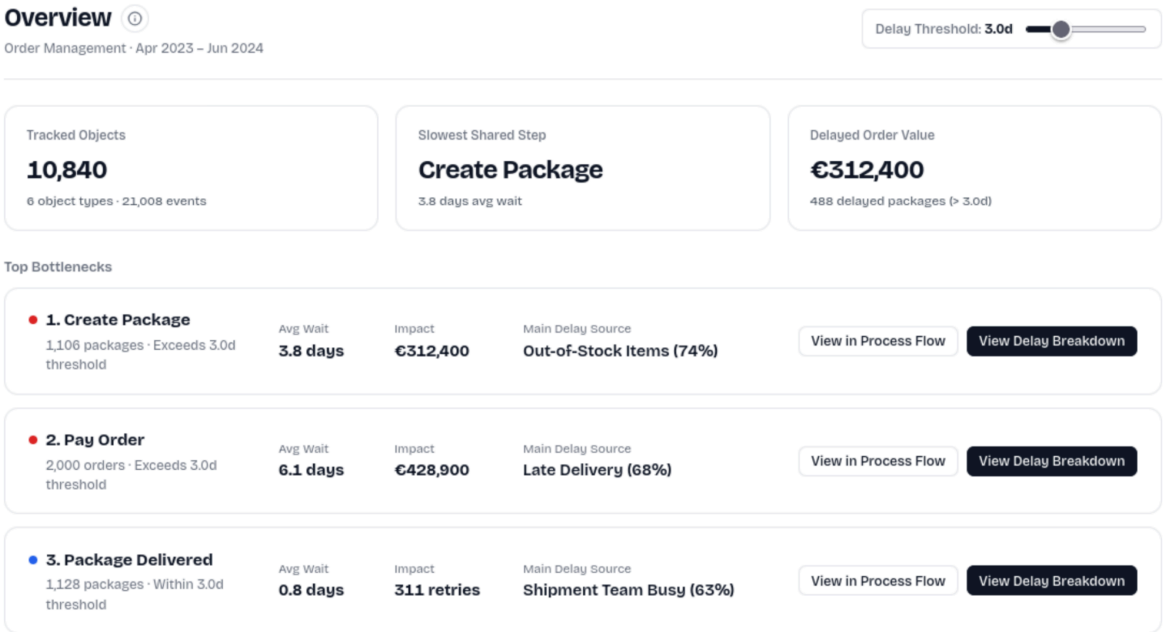


Henri Julius

The app is composed of several subpages that can be accessed via the sidebar on the left.



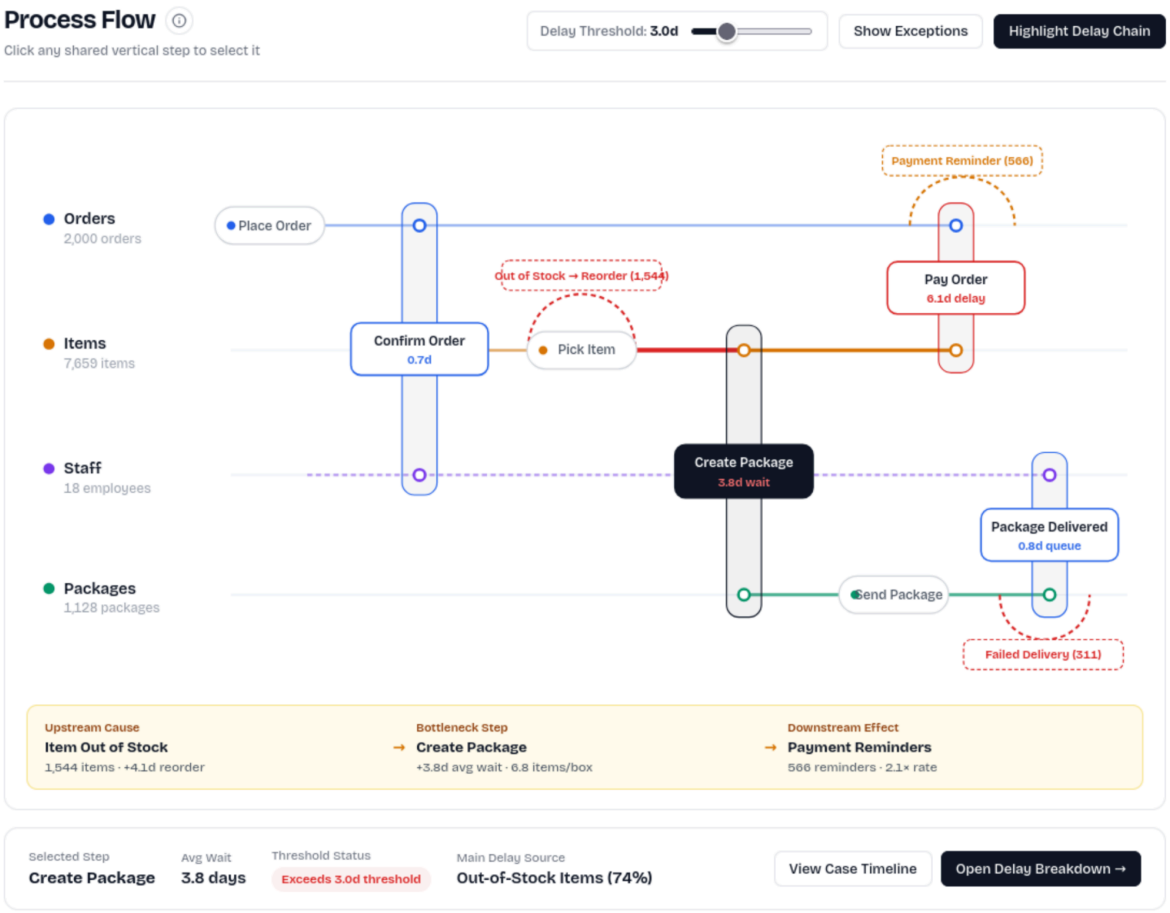
The first subpage is the *Overview*. This shows the three most critical inter-object delays that were extracted from the event log. A delay threshold slider on the top right of the page can be adjusted based on the accepted delay per intersection (activity with relations to multiple types of objects).



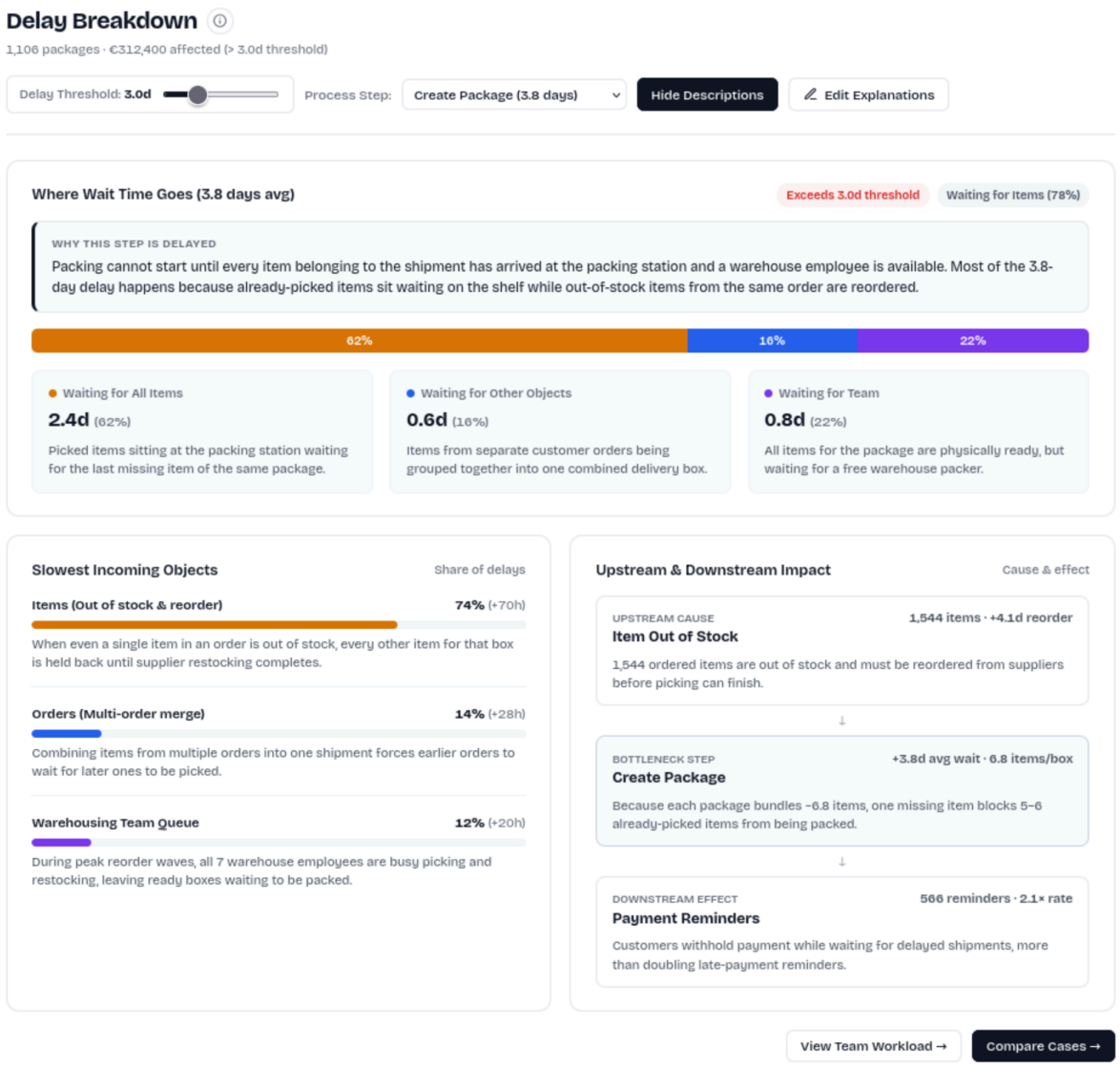
Userguide: M1 Prototype

Henri Julius

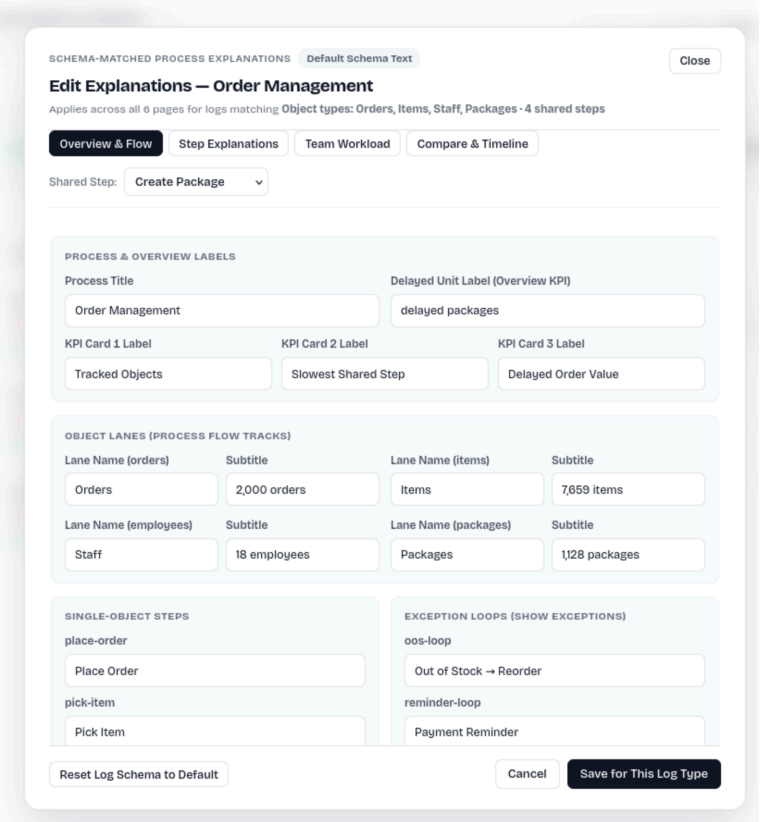
The *Process Flow* shows a metro-map diagram that highlights the intersection points in the process, the object flow connections that delay them the most and optional visualization of delay chains and (uncommon) exceptions.



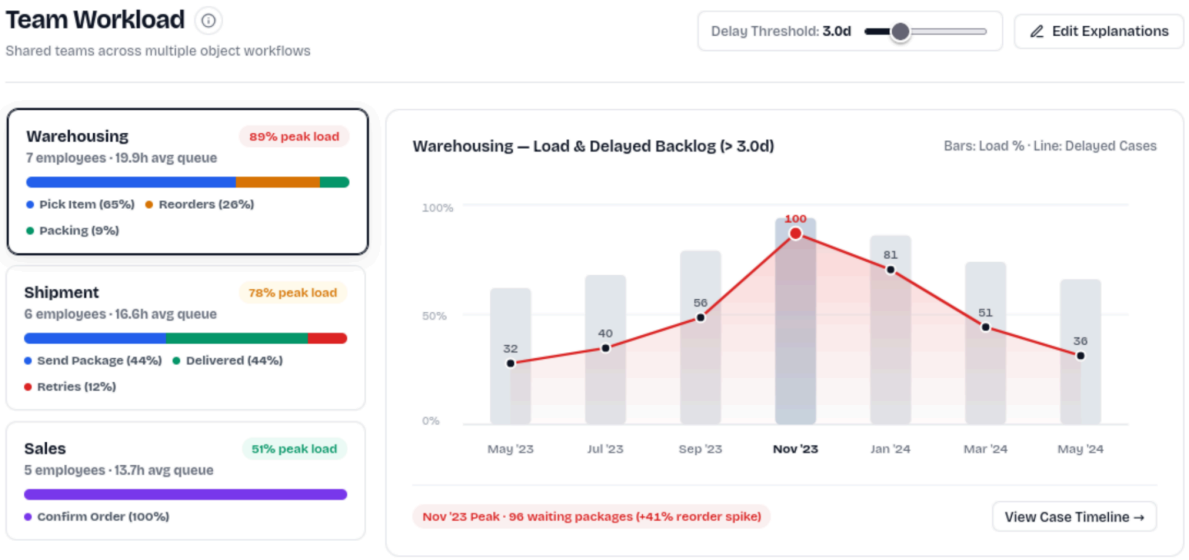
The *Delay Breakdown* is a more practical tool for getting . Textual explanations can be added by analysts for previously known object/event schemas. These textual explanations give non-technical users such as project managers a direct reasoning chain for why the object-centric process gets delayed. In this prototype textual problem descriptions are defined for the two example OCELs.



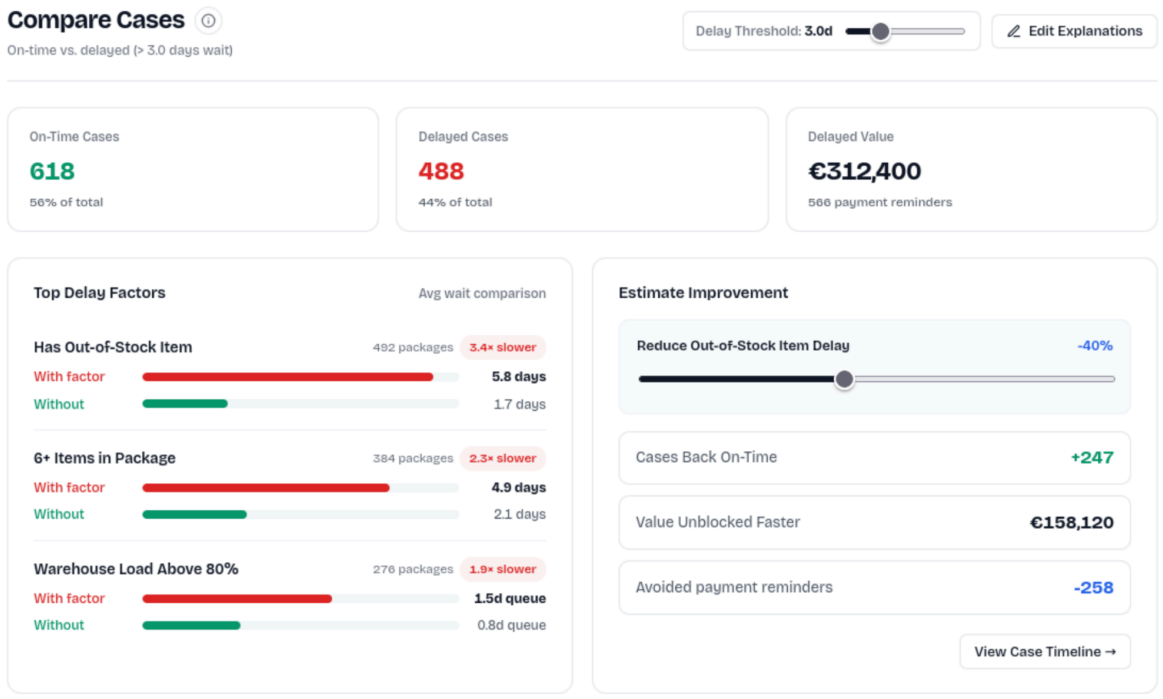
The explanation editor (top right of each page) is a valuable preparation tool which analysts can use to put the process data into context for the non-technical users.



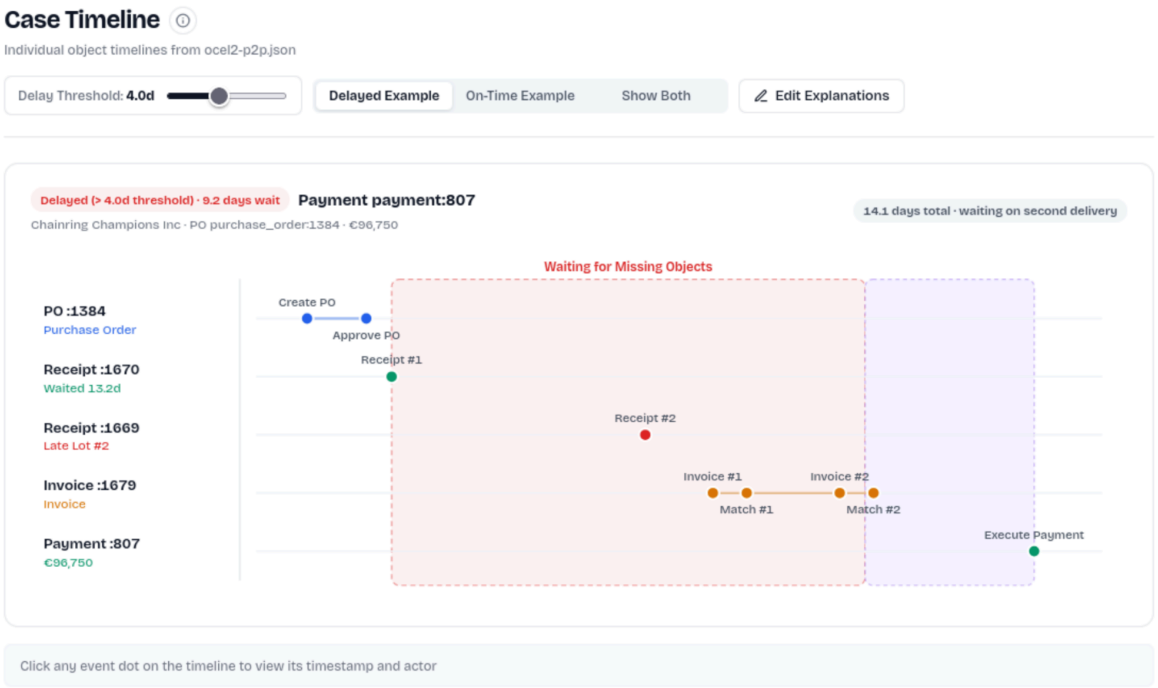
The *Team Workload* page allows team managers and analysts to determine the impact of each involved team on the delays of the overall process. This is crucial for getting a complete image of an inter-object delay, as they are frequently also related to inter-team work being slower than the work in each team.



Compare Cases give an overview of the direct monetary impact of the delays.



Case Timeline shows concrete delayed and non-delayed examples. Seeing which steps are slower and/or repeated in delayed cases compared to non-delayed ones, allows a better analysis of the process specific reasons for the delays.



Future features include:

- Working backend written in Python using libraries such as PM4PY
- Automatic detection of event log schemas and reapplication of known textual descriptions
- Saving and uploading of explanations for specific OCEL schemas
- Call AI models to analyze logs and autonomously create schema descriptions